

Question	Marking Guidance	Mark	Comments
10.1	1. (Large) number/variety of plant species/populations; 2. (More/different) habitats/niches; 3. (Greater) variety/types/sources of food;	3 (3 x AO2)	1. Accept high plant species richness or high index of plant diversity 2. Ignore shelter/homes /environments 3. Ignore 'more food'

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10.2	1. (High/optimum) temperature for enzymes /respiration/nitrification/ammonification; 2. Water/moisture for metabolic/enzyme reactions OR Water for hydrolysis; 3. (Optimum) pH for enzymes/respiration /nitrification/ammonification; 4. (High) oxygen (concentration) for respiration /nitrification /nitrifying bacteria/ saprobionts;	2 max (2 x AO2)	2. Reject 'water for photosynthesis' 4. Accept: saprophytes or saprotrophs for saprobionts

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10.3	1. Protein/amino acids/DNA into ammonium /ammonia; 2. (By) saprobionts; 3. Ammonium/ammonia into nitrite and (then) into nitrate; 4. (By) nitrifying bacteria/microorganisms;	4 (4 x AO1)	1. Accept any named nitrogen containing compound e.g. urea 2. Accept saprophytes or saprotrophs 1 and 3. Accept marks for conversion even if incorrect type of bacteria named as being involved 2 and 4. Reject marks for type of bacteria if linked to incorrect process e.g. nitrite converted to nitrate by saprobionts 2 and 4. Accept correctly named species Ignore nitrogen fixation and denitrification and bacteria involved.

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10.4	1. (Colonisation by) pioneer species; 2. (Pioneers/species/organisms) change the environment/habitat/conditions/factors; 3. (Environment becomes) less hostile for other/new species OR (Environment becomes) more suitable for other/new species OR (Environment becomes) less suitable for previous species; 4. Change/increase in diversity/biodiversity; 5. (To) climax community (of rainforest);	4 max (3 x AO1, 1 x AO2)	1. Accept (as secondary succession) 'Seeds/spores (present)' 2. Accept (as secondary succession) 'Soil/humus or organic matter is already present' 2. Accept example of change e.g. forms soil/humus/organic matter/ nutrients 2. Must convey idea of change being caused by pioneers/species/organisms 3. Accept 'favourable' or 'less harsh' for 'suitable' but ignore 'more hospitable' or 'more habitable' 3. Accept previous species out-competed. 4. Ignore increase in genetic diversity

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10.5	1. Growing crops; 2. Animal farming OR (Land used for) grazing; 3. Ploughing (land); 4. (Applying) weedkillers/herbicides/pesticides; 5. Burning (fields/plants); 6. Building related to farming (barns/greenhouses);	2 max (2 x AO2)	1 Accept specific examples e.g. growing maize and accept 'monoculture' 2. Accept examples e.g. cattle farming 4. Accept insecticides 5. Reject burning fuel